INVESTIGATION LOG

Case File: Murder in SQL City
Lead Investigator: Forensic Data Analyst
Date: November 26, 2025

Step 1: Retrieving the Crime Scene Report

- **The Question:** What are the specific details of the murder crime scene report from January 15, 2018, in SQL City?

- **The SQL Query:**

```
-- Retrieve the official crime scene report for the murder
SELECT
        *
FROM crime_scene_report
WHERE crime_type = 'murder'
AND crime_date = '2018-01-15'
AND city = 'SQL City';
```

- **The Result/Observation**

Data Output Messages Notifications

Showing rows: 1 to 1

	crime_date date	crime_type text	description text	city character varying
1	2018-01-15	murder	Security footage shows that there were 2 witnesses. The first witne...	SQL City

- **The Conclusion:** The crime scene report confirms that it was a murder on 2018-01-15 in SQL city and provides the key lead of two witnesses with their partial address information.

Step 2: Identifying the Two Witnesses

- **The Question:** Who are the two witnesses based on the description from the crime scene report?
- **The SQL Query:**

```
-- Find the first witness (last house on Northwestern Dr) and the second witness (Annabel on Franklin Ave)

-- Find first witness: last house on Northwestern Dr

(SELECT
        person_id,
        name,
        address_number,
        address_street_name
FROM person
WHERE address_street_name  = 'Northwestern Dr'
ORDER BY address_number DESC      --Gets the highest address number(ie last house)
LIMIT 1)

UNION

-- Finds second witness: Annabel on Franklin Ave
(SELECT
        person_id,
        name,
        address_number,
        address_street_name
FROM person
WHERE name ILIKE '%Annabel%'
AND address_street_name = 'Franklin Ave');
```

- **The Result/Observation:**

Data Output Messages Notifications

	person_id integer	name character varying	address_number integer	address_street_name text
1	14887	Morty Schapiro	4919	Northwestern Dr
2	16371	Annabel Miller	103	Franklin Ave

- **The Conclusion:** The two witnesses have been identified as Morty Schapiro (ID: 14887) and Annabel Miller (ID: 16371). They are the only ones who match the description from the crime scene report therefore, their official testimony must be obtained.

Step 3: Gathering Witness Testimony

- **The Question:** What did the witnesses report during their interviews?

- **The SQL Query:**

```
-- Retrieve the interview transcripts for the two witnesses
SELECT
        *
FROM interview
WHERE person_id in(14887, 16371);
```

- **The Result/observation:**

Data Output Messages Notifications

Showing rows: 1 to 2

	person_id integer	transcript text
1	14887	I heard a gunshot and then saw a man run out. He had a "Get Fit Now Gym" bag. The membership number...
2	16371	I saw the murder happen, and I recognized the killer from my gym when I was working out last week on Ja...

- **The conclusion:** The witnesses provided this combined description of the suspect: a male with a "Get Fit Now Gym" gold membership starting with '48Z'(membership_id), who checked in on January 9th, and owns a car with a license plate containing 'H42W'.

Step 4: Identifying the Prime Suspect (The Killer)

- **The Question:** Who matches the description provided by the witnesses?(a gold gym member with an ID starting '48Z' who checked in on Jan 9th and owns a car with 'H42W' in the plate)

- **The SQL Query:**

```
-- identify the killer by cross-referencing the gym membership, check-in date, and driver's license.
SELECT
        p.person_id,
        p.name,
        gm.membership_id,
        dl.plate_number
FROM drivers_license AS dl          --Join Relevant Tables
JOIN person AS p
        ON dl.license_id = p.license_id
JOIN get_fit_now_member AS gm
        ON p.person_id = gm.person_id
JOIN get_fit_now_check_in AS gc
        ON gc.membership_id = gm.membership_id
WHERE gm.membership_id ILIKE '48Z%'             -- Witness 1: Gym bag ID
        AND gm.membership_status = 'gold'       -- Witness 1: gold member
        AND gc.check_in_date = '2018-01-09'     -- Witness 2: saw at gym
```

```
        AND dl.plate_number ILIKE '%H42W%';        -- Witness 1: license plate
```

- **The Result/observation:**

Data Output Messages Notifications

	person_id integer	name character varying	membership_id text	plate_number text
1	67318	Jeremy Bowers	48Z55	0H42W2

- **The conclusion:** Based on the combined witness testimony, we've identified **Jeremy Bowers** as the murderer. The evidence shows he was at the gym when Annabel saw him, and his car plates match what Morty reported.

Step 5: Obtaining the Killer's Testimony

- **The Question**: Does Jeremy Bowers have any additional information to provide (from his interview transcript), such as what his motives are or who may have hired him?
- **The SQL script**:

```
-- Get the interview transcript from the captured killer, Jeremy Bowers(person_id:67318)

SELECT

        *

FROM interview

WHERE person_id = 67318;
```

- **The Result/observation:**

Data Output Messages Notifications

Showing rows: 1 to 1

	person_id integer	transcript text
1	67318	I was hired by a woman with a lot of money. I don't know her name but I know she's around 5'5' (65") or 5'...

- **The conclusion:** Jeremy Bowers confesses to being a hitman. He provides a detailed description of the mastermind behind everything: a wealthy woman, around 5'5" to 5'7" with red hair, who drives a Tesla Model S and attended the SQL Symphony Concert three times in December 2017. This description gives enough detail to trace the mastermind.

Step 6: Identifying the Mastermind

- **The Question**: Who is the wealthy woman matching the description provided by the killer?

- **The script:**

```
-- Find the mastermind based on the description from Jeremy Bowers' testimony
SELECT
        p.person_id,
        p.name,
        dl.gender,
        dl.hair_color,
        dl.car_make,
        dl.car_model
FROM drivers_license AS dl                    --Relevant table Joins
JOIN person AS p
```

```
        ON dl.license_id = p.license_id
JOIN facebook_event_checkin AS fc
        ON fc.person_id = p.person_id
WHERE dl.hair_color= 'red'                    --Mastermind Description
        AND dl.height BETWEEN 65 AND 67
        AND dl.gender = 'female'
        AND dl.car_make = 'Tesla'
        AND dl.car_model = 'Model S'
        AND fc.event_name = 'SQL Symphony Concert'
        AND fc.event_date::text LIKE '2017-12%'
GROUP BY  1,2,3,4,5,6
HAVING COUNT(fc.event_id)=3;                  -- Attended 3 times in December
```

- **The Result/observation:**

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1

	person_id integer	name character varying	gender text	hair_color text	car_make text	car_model text
1	99716	Miranda Priestly	female	red	Tesla	Model S

- **The Conclusion:** The mastermind who hired Jeremy Bowers is Miranda Priestly. She fits the exact physical description, drives the specified car, and is confirmed to have attended the SQL Symphony Concert three times in December 2017.

THE SINGLE QUERY:

```
---The queries above can be combined into one query using the CTE below:



--For the Culprit

WITH culprit AS (

        SELECT

            p.name,

            p.person_id,

            p.license_id,

            d.plate_number,

            g.membership_status,

            g.membership_id,

            gc.check_in_date

        FROM drivers_license AS d                    --Joins relevant tables

        JOIN person AS p

            ON p.license_id = d.license_id

        JOIN get_fit_now_member AS g

            ON p.person_id = g.person_id

        JOIN get_fit_now_check_in AS gc

            ON gc.membership_id = g.membership_id

        WHERE d.plate_number ILIKE '%H42W%'                  -- Witness 1: license plate

          AND g.membership_id ILIKE '%48Z%'                  -- Witness 1: Gym bag ID
```

```
        AND gc.check_in_date = '2018-01-09'                    -- Witness 2: saw at gym
        ),


--For the Mastermind
mastermind AS (
        SELECT
                p.person_id,
                p.name,
                dl.gender,
                dl.hair_color,
                dl.car_make,
                dl.car_model
        FROM drivers_license AS dl                      --Relevant table Joins
        JOIN person AS p
                ON dl.license_id = p.license_id
        JOIN facebook_event_checkin AS fc
                ON fc.person_id = p.person_id
        WHERE dl.hair_color= 'red'                      --Mastermind Description
                AND dl.height BETWEEN 65 AND 67
                AND dl.gender = 'female'
                AND dl.car_make = 'Tesla'
                AND dl.car_model = 'Model S'
```

```
                AND fc.event_name = 'SQL Symphony Concert'

                AND fc.event_date::text LIKE '2017-12%'

        GROUP BY 1,2,3,4,5,6

        HAVING COUNT(fc.event_id) = 3

        )



--Displaying the Murderer

SELECT

        name,

        'culprit' AS crime

FROM culprit

        UNION                   --Helps combine the results from the two queries into one table.

--Displaying the MasterMind

SELECT

        name,

        'mastermind' AS crime

FROM mastermind;
```

- **The Result/observation:**

Data Output Messages Notifications

	name character varying	crime text
1	Jeremy Bowers	culprit
2	Miranda Priestly	mastermind

FINAL CONCLUSION

The conclusion of this case is that Jeremy Bowers committed the murder on January 15, 2018, but he was just a hitman. The real mastermind was Miranda Priestly , a wealthy woman who paid him to do her dirty work. While we may never know exactly why she ordered the hit, she checked all the boxes: red hair, Tesla Model S, and was spotted at that SQL Symphony Concert three times in December. Therefore, Jeremy Bowers is the suspect of murder, and Miranda Priestly, also for ordering it. That is justice served, case closed!